

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$ by changing into polar integration. 5
- b) Evaluate: $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$ 5
- c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2 + y^2 = 4$ and $z + y = 3$. 5
2. a) i) Find the Laplace Transform of: $\frac{\sin^2 t}{t}$ 8
ii) Find the inverse Laplace Transform of: $\frac{w}{s^2(s^2 + w^2)}$
- b) Solve the initial value problem: $y'' + 2y' - 3y = 6e^{-2t}$ where, $y(0) = 2$, $y'(0) = -14$ using Laplace Transform. 7
3. a) Solve $y'' + y = 0$ by using power series method. 7
- b) Derive Rodrigue's formula for Legendre polynomials. 8
- OR
- Define Bessel's differential equation of order n . Find its complete solution.
4. a) Find the directional derivatives of $f = xy^2 + yz^3$ at $P(2, -1, 1)$ along the direction of the normal to the surface $x \log z - y^2 + 4 = 0$ at $(-1, 2, 1)$. 5
- b) If $\phi = \log(x^2 + y^2 + z^2)$, find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$. 5
- c) Show that the value under integral sign is exact and then evaluate integral, 5

(1,1,1)

$$\int [yz \sinh (xz) dx + \cosh (xz) dy + xy \sinh (xz) dz].$$

(0,2,3)

5. a) Let S be the hemi-sphere given by $z = \sqrt{9 - x^2 - y^2}$ and $\vec{F} = (3x, 3y, z)$. Find the flux of $\vec{F}$ through S. 7
- b) State Gauss divergence theorem. Evaluate $\iiint_S \vec{F} \cdot \vec{n} dA$ where $\vec{F} = (y^3, x^3, z^3)$ and S is the cylinder $x^2 + 4y^2 = 1$ $0 < z < 5, x > 0, y > 0$ by using it. 8

OR

State Stoke's theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y^3, 0, x^3)$ and C is the boundary of the triangle with vertices are given by the vectors (1, 0, 0), (0, 1, 0), (0, 0, 1).

6. a) Find the fourier series of $f(x) = x + |x|$ for $-\pi < x < \pi$. 7
- b) Find the fourier cosine series of $f(x) = x \sin x, 0 < x < \pi$ then show that $\frac{\pi-2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \dots$. 8
7. Attempt **any two** questions. 2×5
- a) Solve: $u_t + 2uu_x = 0, u(x, 0) = e^{-x}$.
- b) Derive the one-dimensional traffic flow model using conservation law.
- c) Find the angle between the surfaces $xy^2z - 3x - z^2 = 0$ and $3x^2 + 2z = 1 + y^2$ at (1, -2, 1)